MICROSOFT EXCEL

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1. INTRODUCTION TO MS EXCEL

What is MS Excel

MS Excel is a spreadsheet software made by Microsoft.

Excel is used for:

- Data Entry
- Calculations
- Reports
- Charts
- Salary Sheet
- Attendance
- Billing
- Data Analysis
- Dashboard
- Automation

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2. EXCEL INTERFACE

Main Parts of Excel Window

Part	Meaning
Title Bar	Shows file name
Ribbon	Contains tools and commands
Workbook	Excel file
Worksheet	Sheet inside workbook
Row	Horizontal line (1,2,3...)
Column	Vertical line (A,B,C...)
Cell	Intersection of row and column
Formula Bar	Shows formula
Name Box	Shows cell address

Example:

- A1 = Column A + Row 1

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3. WORKBOOK AND WORKSHEET

Workbook

Entire Excel file.

Example:

- Salary.xlsx

Worksheet

Single sheet inside workbook.

Example:

- Sheet1
- Sheet2

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4. DATA TYPES IN EXCEL

Type	Example
Text	Vijay
Number	100
Date	12/05/2026
Time	10:30 AM
Formula	=A1+B1
Percentage	50%

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5. HOW TO ENTER DATA

1. Click cell
2. Type data
3. Press Enter

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6. BASIC OPERATIONS

Operation	Formula
Addition	=A1+B1
Subtraction	=A1-B1
Multiplication	=A1*B1
Division	=A1/B1

Example:

A	B
10	5

Formula:

- =A1+B1 → 15

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7. CELL REFERENCES

1. Relative Reference

Formula changes automatically.

Example:
= A1+B1

2. Absolute Reference

Fixed cell reference.

Example:
= A1

Shortcut:
F4

3. Mixed Reference

Example:

- $A1
- A$1

Important:

- A1 → Absolute Reference
- A$1 → Row Fixed
- $A1 → Column Fixed

F4 Key Magic:

Press	Result
1st	A1
2nd	A$1
3rd	$A1
4th	A1

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8. AUTO FILL

Used to copy formula automatically.

Example:
Drag small square from cell corner.

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9. BASIC FORMATTING

Formatting Options

- Bold
- Italic
- Underline
- Font Size
- Font Color
- Fill Color
- Border
- Alignment

Important Shortcut Keys:

Shortcut	Use
Ctrl + B	Bold
Ctrl + I	Italic
Ctrl + U	Underline
Ctrl + 1	Format Cells
Alt + H + O + I	AutoFit Column

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10. NUMBER FORMATTING

Format	Example
Currency	₹100
Percentage	50%
Date	01-01-2026
Time	10:00 AM

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11. IMPORTANT SHORTCUT KEYS

Shortcut	Work
Ctrl + N	New Workbook
Ctrl + O	Open
Ctrl + S	Save
Ctrl + P	Print
Ctrl + C	Copy
Ctrl + V	Paste
Ctrl + X	Cut
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + A	Select All
Ctrl + F	Find
Ctrl + H	Replace
Ctrl + Arrow Key	Move to last data
Ctrl + Shift + L	Filter
F2	Edit Cell
F4	Repeat / Absolute Reference
Alt + =	AutoSum
Ctrl + ;	Current Date
Ctrl + Shift + ;	Current Time

12. BASIC FORMULAS

SUM

Adds numbers.

Example:
=SUM(A1:A5)

AVERAGE

Finds average.

Example:
=AVERAGE(A1:A5)

MAX

Largest number.

Example:
=MAX(A1:A5)

MIN

Smallest number.

Example:
=MIN(A1:A5)

COUNT

Counts numbers only.

Example:
=COUNT(A1:A5)

COUNTA

Counts non-empty cells.

Example:
=COUNTA(A1:A5)

ROUND

Rounds number.

Example:
=ROUND(12.567,2)

Output:
12.57

---

13. TEXT FUNCTIONS

LEFT

Extract left side text.

Example:
=LEFT("VIJAY",2)

Output:
VI

RIGHT

Example:
=RIGHT("VIJAY",2)

Output:
AY

MID

Example:
=MID("VIJAY",2,3)

Output:
IJA

LEN

Counts characters.

Example:
=LEN("VIJAY")

Output:
5

UPPER

Example:
=UPPER("vijay")

Output:
VIJAY

LOWER

Example:
=LOWER("VIJAY")

Output:
vijay

PROPER

Example:
=PROPER("vijay kumar")

Output:
Vijay Kumar

CONCAT / CONCATENATE

Joins text.

Example:
=A1&" "&B1

TRIM

Removes extra spaces.

Example:
=TRIM(A1)

FIND

Finds position of text.

Example:
=FIND("a","Vijay")

Output:
4

Important:

- FIND function Case-Sensitive होता है।
- Small letter और Capital letter में फर्क समझता है।

Example:
=FIND("V","Vijay") → 1

=FIND("v","Vijay") → Error

Interview Point:

- FIND = Case-Sensitive
- SEARCH = Not Case-Sensitive

SUBSTITUTE

Replace text.

Example:
=SUBSTITUTE(A1,"old","new")

---

14. DATE & TIME FUNCTIONS

TODAY

Current date.

Example:
=TODAY()

NOW

Current date and time.

Example:
=NOW()

DAY

Example:
=DAY(A1)

MONTH

Example:
=MONTH(A1)

YEAR

Example:
=YEAR(A1)

DATEDIF

Difference between dates.

Example:
=DATEDIF(A1,B1,"Y")

TEXT

Formats date/text.

Example:
=TEXT(A1,"DD-MM-YYYY")

---

15. LOGICAL FUNCTIONS

IF FUNCTION

Condition checking.

Correct Syntax: =IF(A1>50,"Pass","Fail")

Example: =IF(A1>=40,"Pass","Fail")

Real Life Example:
अगर student के marks 40 से ज्यादा हैं तो Pass नहीं तो Fail।

NESTED IF

Multiple conditions.

Example:
=IF(A1>=80,"A",IF(A1>=60,"B","C"))

AND FUNCTION

All conditions true.

Example:
=AND(A1>50,B1>50)

OR FUNCTION

Any condition true.

Example:
=OR(A1>50,B1>50)

NOT FUNCTION

Opposite result.

Example:
=NOT(A1>50)

IFS FUNCTION

Multiple conditions easier.

Correct Example:
=IFS(A1>=90,"A+",A1>=80,"A",TRUE,"Below 80")

Important:

- अगर कोई condition match नहीं होती तो IFS function #N/A Error देता है।
- इसलिए अंत में TRUE,"Default Value" लगाना best practice है।

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16. LOOKUP FUNCTIONS

VLOOKUP

Vertical search.

Syntax:
=VLOOKUP(lookup_value,table_array,col_index,FALSE)

Example:
=VLOOKUP(101,A2:D10,2,FALSE)

Real Life Example:
Find employee name using ID.

HLOOKUP

Horizontal lookup.

Example:
=HLOOKUP(101,A1:G5,2,FALSE)

XLOOKUP (Advanced & Important)

Modern lookup formula.

Example:
=XLOOKUP(E2,A2:A10,B2:B10)

Advantages:

- Easier than VLOOKUP
- Left lookup possible
- More powerful

INDEX + MATCH

Advanced lookup formula.

Correct Formula:
=INDEX(B2:B10,MATCH(E2,A2:A10,0))

Interview Important Topic.

Example Explanation:

- MATCH function position ढूंढता है।
- INDEX function उस position की value return करता है।

Real Life Example:
Employee ID से Employee Name निकालना।

VLOOKUP vs XLOOKUP (INTERVIEW IMPORTANT)

VLOOKUP	XLOOKUP
केवल Right Side lookup	Left और Right दोनों lookup
Column number देना पड़ता है	Direct range selection
Less flexible	More flexible
Error handling अलग से	Better error handling
Old function	Modern function

Interview Answer:

"XLOOKUP ज्यादा powerful है क्योंकि यह left lookup कर सकता है, ज्यादा flexible है और modern Excel function है।"

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17. CONDITIONAL FUNCTIONS

COUNTIF

Count with condition.

Example:

=COUNTIF(A1:A10,">50")

COUNTIFS

Multiple conditions.

Example:
=COUNTIFS(A1:A10,">50",B1:B10,"Pass")

SUMIF

Conditional sum.

Example:
=SUMIF(A1:A10,"Apple",B1:B10)

SUMIFS

Multiple condition sum.

Example:
=SUMIFS(C1:C10,A1:A10,"Apple",B1:B10,"North")

AVERAGEIF

Conditional average.

Example:
=AVERAGEIF(A1:A10,">50")

---

18. ERROR HANDLING FUNCTIONS

IFERROR

Handles errors.

Example:
=IFERROR(A1/B1,"Error")

ISERROR

Checks error.

Example:
=ISERROR(A1/B1)

COMMON EXCEL ERRORS

Error	Meaning
#DIV/0!	Divide by zero
#NAME?	Wrong formula name
#VALUE!	Wrong data type
#REF!	Invalid reference
#N/A	Value not found

#REF! ERROR

Meaning:
Invalid Cell Reference

Main Reason:
जब formula में इस्तेमाल की गई row या column delete हो जाए।

Example:
=A1+B1

अगर B column delete कर दिया जाए तो:
#REF! Error आ सकता है।

Interview Answer:
"#REF! error तब आता है जब formula की referenced cell/row/column delete हो जाती है।"

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19. DATA SORTING

Ascending

A to Z

Descending

Z to A

Shortcut:
Alt + D + S

20. FILTER

Used to filter data.

Shortcut:
Ctrl + Shift + L

Example:
Show only Sales department employees.

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21. CONDITIONAL FORMATTING

Highlights data automatically.

Examples:

- Duplicate values
- Top 10 values
- Color scales

Real Life Example:
Highlight students below 40 marks.

---

22. DATA VALIDATION

Restricts wrong input.

Example:
Allow only:

- Numbers
- Date
- Drop-down list

Real Life Example:
Gender dropdown:
Male/Female

---

23. REMOVE DUPLICATES

Used to remove repeated data.

Location:
Data → Remove Duplicates

---

24. TEXT TO COLUMNS

Splits data.

Example:
"Vijay Kumar"

Split into:

- Vijay

- Kumar

---

25. FLASH FILL

Automatically detects pattern.

Shortcut:
Ctrl + E

Example:
Convert:
vijaykumar977129@gmail.com
to
VIJAY

---

26. TABLES IN EXCEL

Shortcut:
Ctrl + T

Advantages:

- Auto filter
- Better formatting
- Dynamic formulas

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27. CHARTS IN EXCEL

Types of Charts

Chart	Use
Column Chart	Compare values
Bar Chart	Horizontal comparison
Pie Chart	Percentage
Line Chart	Trends
Area Chart	Growth
Scatter Plot	Relationship

Real Life Example:
Monthly sales chart.

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28. PIVOT TABLE (VERY IMPORTANT)

Used for data analysis.

Steps

1. Select data
2. Insert
3. Pivot Table
4. Drag fields

Example:
Find total sales by department.

Interview Very Important.

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29. PIVOT CHART

Chart from Pivot Table.

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30. SLICER

Used for filtering Pivot Tables visually.

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31. FREEZE PANES

Locks rows/columns.

Example:
Freeze heading row.

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32. PAGE LAYOUT

Used before printing.

Options:

- Margin
- Orientation
- Size

- Print Area

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33. PRINTING OPTIONS

Shortcut:
Ctrl + P

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34. PROTECT SHEET

Prevents editing.

Steps:
Review → Protect Sheet

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35. HIDE & UNHIDE

Used to hide:

- Rows
- Columns
- Sheets

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36. FIND & REPLACE

Shortcut:
Ctrl + F
Ctrl + H

37. GO TO SPECIAL

Shortcut:
F5

Used for:

- Blank cells
- Formulas
- Constants

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38. NAMED RANGE

Give name to range.

Example:
SalesData

Formula:
=SUM(SalesData)

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39. SPARKLINES

Mini charts inside cell.

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40. WHAT IF ANALYSIS

Goal Seek

Find required value.

Example:
How many marks needed to get 90%.

Data Table

Multiple calculations.

Scenario Manager

Different scenarios.

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41. ADVANCED FILTER

Advanced filtering conditions.

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42. POWER QUERY (ADVANCED)

Used for:

- Cleaning data
- Combining files
- Transforming data

Very Important for Data Analyst.

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43. POWER PIVOT

Advanced data modeling.

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44. DASHBOARD IN EXCEL

Dashboard means interactive report.

Uses:

- Charts
- KPIs
- Slicers
- Pivot Tables

Real Life Example:
Company sales dashboard.

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45. MACROS (ADVANCED)

Macro automates tasks.

Record Macro

View → Macros

Shortcut:
Alt + F8

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46. VBA (VISUAL BASIC FOR APPLICATIONS)

Programming language in Excel.

Used for:

- Automation
- User Forms
- Buttons

Example:

```
Sub Hello()
MsgBox "Hello"
End Sub
```

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47. IMPORTANT INTERVIEW QUESTIONS

Basic Questions

1. What is Excel?
2. Difference between Workbook andWorksheet
3. What is a Cell?
4. What is Formula?
5. Difference between Relative and Absolute Reference?

Formula Questions

1. Difference between COUNT and COUNTA?
2. Difference between VLOOKUP and XLOOKUP?
3. Difference between SUMIF and SUMIFS?
4. What is IFERROR?

Advanced Questions

1. What is Pivot Table?
2. What is Macro?
3. What is Power Query?
4. What is Dashboard?
5. What is Conditional Formatting?

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48. PRACTICE PROJECTS

Beginner Projects

1. Student Marksheet
2. Attendance Sheet
3. Expense Tracker
4. Salary Sheet

Intermediate Projects

1. Sales Report
2. Inventory Management
3. Employee Database

Advanced Projects

1. Interactive Dashboard
2. Automated Invoice
3. Data Analysis Report

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49. EXCEL BEST PRACTICES

1. Use Tables
2. Use Proper Headers
3. Avoid Merged Cells
4. Keep Backup
5. Use Named Ranges
6. Use Data Validation
7. Keep formulas simple

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50. REAL LIFE USES OF EXCEL

Field	Use
Banking	Reports
Office	Attendance
School	Result
Business	Billing
Data Analyst	Analysis
HR	Salary

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51. IMPORTANT EXCEL FORMULAS LIST

Must Learn Formulas

- SUM
- AVERAGE
- MAX
- MIN
- COUNT
- COUNTA
- IF
- AND
- OR

- VLOOKUP
- XLOOKUP
- INDEX
- MATCH
- SUMIF
- SUMIFS
- COUNTIF
- COUNTIFS
- IFERROR
- LEFT
- RIGHT
- MID
- LEN
- CONCAT
- TODAY
- NOW
- TEXT
- ROUND

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52. EXCEL LEARNING ROADMAP

Step 1

Learn:

- Interface
- Formatting
- Basic formulas

Step 2

Learn:

- IF
- Lookup formulas
- Charts
- Filter

Step 3

Learn:

- Pivot Table
- Conditional Formatting
- Data Validation

Step 4

Learn:

- Dashboard
- Power Query
- Macros
- VBA

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53. FINAL PRACTICE ADVICE

Daily Practice:

- 30 min formulas
- 30 min reports
- 30 min dashboard practice

Best Practice Method:

1. Learn formula
2. Create small example
3. Use in project
4. Repeat daily

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54. MOST IMPORTANT ADVANCED TOPICS

These topics are highly important for jobs/interviews:

- XLOOKUP
- INDEX + MATCH
- Pivot Table
- Pivot Chart
- Dashboard
- Power Query
- Conditional Formatting
- Data Validation
- IFERROR
- Macros
- VBA
- Dynamic Charts
- Slicers

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55. EXCEL FILE EXTENSIONS

Extension	Meaning
.xlsx	Normal Excel File
.xls	Old Excel File
.csv	Comma Separated
.xlsm	Macro Enabled

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56. DIFFERENCE BETWEEN CSV AND XLSX

CSV	XLSX
Simple text file	Excel workbook
No formulas	Supports formulas
No formatting	Supports formatting

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57. DYNAMIC ARRAY FUNCTIONS (ADVANCED)

FILTER

Example:
=FILTER(A2:C10,B2:B10="Sales")

SORT

Example:
=SORT(A2:A10)

UNIQUE

Example:
=UNIQUE(A2:A10)

SEQUENCE

Example:
=SEQUENCE(10)

---

58. FINANCIAL FUNCTIONS

PMT

Loan EMI calculation.

Example:
=PMT(10%/12,60,-500000)

FV

Future value.

Example:
=FV(10%,5,-1000)

59. STATISTICAL FUNCTIONS

MEDIAN

Example:
=MEDIAN(A1:A10)

MODE

Example:
=MODE(A1:A10)

STDEV

Example:
=STDEV(A1:A10)

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60. IMPORTANT EXCEL TIPS

1. Use Freeze Panes for big data
2. Use Pivot Table for reports
3. Use IFERROR to avoid ugly errors
4. Use Tables for dynamic data
5. Learn shortcuts gradually

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61. EXCEL FOR DATA ANALYST

Must Learn:

- Power Query
- Pivot Table
- XLOOKUP
- INDEX MATCH
- Charts
- Dashboard
- Power Pivot

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62. EXCEL FOR OFFICE JOB

Must Learn:

- Formatting
- Basic formulas
- Filters
- Sort
- Print setup
- Charts
- VLOOKUP

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63. FINAL INTERVIEW PREPARATION

If you know these properly:

- VLOOKUP
- XLOOKUP
- Pivot Table
- IF Formula
- Conditional Formatting
- Charts
- Filters
- Data Validation

Then you can clear most basic Excel interviews.

END OF COMPLETE EXCEL NOTES